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IMPLEMENTATION COMPLETION AND RESULTS REPORT

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ON A

CREDIT

IN THE AMOUNT OF SDR 70.6 MILLION

(US\$100 MILLION EQUIVALENT)

TO THE

PEOPLE'S REPUBLIC OF BANGLADESH

FOR THE

COLLEGE EDUCATION DEVELOPMENT PROJECT

(June 10, 2024)

Education Global Practice
South Asia Region

CURRENCY EQUIVALENTS
(Exchange Rate Effective March 30, 2024)

Currency Unit = Bangladeshi Taka (BDT)

BDT 117 = US\$1

US\$1.32 = SDR 1

FISCAL YEAR
July 1 – June 30

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ABBREVIATIONS AND ACRONYMS

BdREN	Bangladesh Research and Education Network
BIDS	Bangladesh Institute of Development Studies
BOU	Bangladesh Open University
CEDP	College Education Development Project
DLI	Disbursement-Linked Indicator
DLR	Disbursement-Linked Result
DPP	Development Project Proposal
DSHE	Directorate of Secondary and Higher Education
ECPD	Enhanced Continuous Professional Development
e-GP	Electronic Government Procurement
EOP	End-of-Project
FGD	Focus Group Discussion
GDP	Gross Domestic Product
GoB	Government of Bangladesh
HEQEP	Higher Education Quality Enhancement Project
ICT	Information and Communication Technology
IDG	Institutional Development Grant
IDP	Institutional Development Plan
IRI	Intermediate Results Indicator
IRR	Internal Rate of Return
KII	Key Informant Interview
LMIS	Learning Management Information System
M&E	Monitoring and Evaluation
MOE	Ministry of Education
MTR	Midterm Review
NAEM	National Academy for Education Management
NPV	Net Present Value
NSP	National Strategic Plan
NTRCA	Nongovernment Teachers' Registration and Certification Authority
NTSC	Nongovernment Teachers' Selection Commission
NU	National University
PAD	Project Appraisal Document
PDO	Project Development Objective
PMIS	Project Management Information System
PMU	Project Management Unit
SEVC	Small Ethnic and Vulnerable Communities
SHED	Secondary and Higher Education Division
TOC	Theory of Change
UNM	University of Nottingham Malaysia

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DATA SHEET

BASIC INFORMATION

Product Information

Project ID	Project Name
P154577	College Education Development Project
Country	Financing Instrument
Bangladesh	Investment Project Financing
Original EA Category	Revised EA Category
Partial Assessment (B)	Partial Assessment (B)

Organizations

Borrower	Implementing Agency
People's Republic of Bangladesh	Ministry of Education, National University, University Grants Commission

Project Development Objective (PDO)

Original PDO

The project development objectives are to strengthen the strategic planning and management capacity of the college education subsector and to improve the teaching and learning environment of participating colleges.

**FINANCING**

	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Financing			
IDA-58410	100,000,000	93,579,642	89,474,731
Total	100,000,000	93,579,642	89,474,731
Non-World Bank Financing			
Borrower/Recipient	30,000,000	0	0
Total	30,000,000	0	0
Total Project Cost	130,000,000	93,579,642	89,474,731

KEY DATES

Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
03-Jun-2016	13-Oct-2016	14-Jan-2019	30-Jun-2022	31-Dec-2023

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US\$M)	Key Revisions
01-May-2019	25.08	Change in Results Framework Change in Loan Closing Date(s) Change in Disbursements Arrangements Change in Implementation Schedule
14-Jun-2021	42.01	Change in Results Framework Change in Components and Cost
10-May-2023	65.12	Change in Results Framework Change in Components and Cost Reallocation between Disbursement Categories
05-Dec-2023	76.25	Change in Results Framework Change in Components and Cost Cancellation of Financing Reallocation between Disbursement Categories

KEY RATINGS

Outcome	Bank Performance	M&E Quality
Satisfactory	Satisfactory	Substantial



RATINGS OF PROJECT PERFORMANCE IN ISRs

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	09-Sep-2016	Satisfactory	Satisfactory	0
02	15-Feb-2017	Satisfactory	Satisfactory	0
03	08-Sep-2017	Satisfactory	Moderately Satisfactory	20.41
04	04-Mar-2018	Satisfactory	Moderately Unsatisfactory	20.41
05	23-Aug-2018	Moderately Unsatisfactory	Moderately Unsatisfactory	20.41
06	19-Feb-2019	Moderately Unsatisfactory	Moderately Satisfactory	25.08
07	23-Jul-2019	Moderately Satisfactory	Moderately Satisfactory	28.99
08	14-Nov-2019	Moderately Satisfactory	Satisfactory	29.52
09	27-May-2020	Moderately Satisfactory	Satisfactory	37.99
10	01-Nov-2020	Moderately Satisfactory	Moderately Satisfactory	37.99
11	05-May-2021	Satisfactory	Moderately Satisfactory	42.01
12	10-Nov-2021	Satisfactory	Satisfactory	43.11
13	11-May-2022	Satisfactory	Satisfactory	50.02
14	08-Aug-2022	Satisfactory	Satisfactory	59.47
15	05-Feb-2023	Satisfactory	Moderately Satisfactory	59.47
16	19-Jun-2023	Satisfactory	Moderately Satisfactory	65.12
17	27-Nov-2023	Satisfactory	Moderately Satisfactory	76.25

SECTORS AND THEMES

Sectors

Major Sector/Sector

(%)

Education

100

Public Administration - Education

10

Tertiary Education

90



Themes

Major Theme/ Theme (Level 2)/ Theme (Level 3) (%)

Private Sector Development 100

Jobs 100

Human Development and Gender 100

Education 100

Access to Education 25

Science and Technology 25

Teachers 25

Standards, Curriculum and Textbooks 25

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I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

- 1. At the time of appraisal of the College Education Development Project (CEDP), in June 2016, Bangladesh had graduated from a low-income to a lower-middle-income country, with impressive economic growth and poverty reduction.** The annual gross domestic product (GDP) growth rate increased from 5.57 percent in 2010 to 7.11 percent in 2016, with a GDP per capita of US\$1,660 in 2016.¹ Human development outcomes also improved. The maternal mortality rate fell by 34 percentage points from 2010 to 2016. The net primary enrollment increased from 66.5 percent in 2005 to 97.3 percent in 2013. Enrollments also increased at the secondary and tertiary education levels.² Gender parity was achieved in primary and secondary education levels.
- 2. The Government of Bangladesh (GoB) recognized the need to enhance the quality and relevance of the higher education sector to sustain the country's economic progress.** The Government planned to improve the National University (NU)-affiliated college system, which included over 1,700 tertiary colleges and accounted for nearly two-thirds of tertiary enrollment (approximately 1.6 million college students). The colleges catered to students mostly from the rural areas and from low- and middle-income families. Students in rural colleges were more likely to be from poorer families compared to those in urban colleges and universities.³ Female student participation accounted for 45 percent of total enrollments in the colleges. The high number of colleges and their extensive network across Bangladesh therefore placed the NU-affiliated colleges in a strategic position to improve gender parity in the tertiary education sector and improve economic outcomes for less advantaged students.
- 3. However, this subsector faced significant challenges.** First, inadequate accountability and capacity at both national and institutional levels hindered effective planning, governance, and management, with no long-term strategic planning. Second, low quality and relevance were issues due to a shortage of qualified teaching staff, lack of incentives, and accountability mechanisms. Most college teachers lacked updated training in their fields and pedagogical skills or academic qualifications beyond a Master's degree. Insufficient teaching-learning facilities adversely affected the classroom environment, and the NU curriculum lacked market relevance. Additionally, the NU college system did not have proper quality assurance mechanisms. Third, inadequate financing led to low development investments in the sector. Financing from the Ministry of Education (MOE) to government colleges, as share of total funding in higher education, declined from 7.0 percent in 2009 to 4.3 percent in 2015. The major share of government college funds was spent on recurrent expenditures, leaving a small amount for academic development and quality enhancement activities. Fourth, the sector suffered from limited access, with enrollment in tertiary education remaining low (gross enrollment rate: 13 percent), especially for females and compared with other countries in the region.
- 4. Rationale for World Bank involvement.** The World Bank had been a long-standing and trusted development partner through various operations in the school and higher education and skill development subsectors in Bangladesh. The World Bank was financing the ongoing Higher Education

¹ Bangladesh Economic Review 2023, Finance Division, GoB.

² Bangladesh Bureau of Educational Information and Statistics (BANBEIS), Education Statistics 2022.

³ World Bank. 2014. *A Study on National University and its Affiliated Colleges*. Report No. 65. South Asia Human Development Sector, Washington DC: World Bank.



Quality Enhancement Project (P106216), and the CEDP was intended to bring fundamental changes at the systemic and institutional levels within the NU and its affiliated colleges system.

Theory of Change (Results Chain)

5. The Theory of Change (ToC) in table 1 has been developed retroactively for this Implementation Completion and Results Report (ICR), based on the Project Appraisal Document (PAD) and actual implementation. The ToC assumes that incentives to support system-level reforms, use new teaching skills and modern equipment, and provide digital infrastructure strengthened strategic planning and management capacity in the college subsector and improved teaching and learning environment in participating colleges.

Table 1. CEDP ToC

Activities	Outputs	Intermediate Outcome	Project Outcome	Long-Term Outcome
Component 1: Strengthening Strategic Planning and Management Capacity: - Develop strategic plan for college subsector - Draft Development Project Proposal (DPP) - Recruit teachers - Conduct teacher needs assessment - Establish Nongovernment Teachers' Selection Commission (NTSC)	- Six background research studies prepared. - National Strategic Plan (NSP) for higher education colleges 2023–2031 approved. - DPP approved. Investment Plan for Higher Education Colleges approved. 3,500 government teachers recruited. - Teacher needs assessment for government colleges completed. Amended NTRCA Act (DLI3)	- Sector strategy and action plan developed, adopted, and the initial implementation started (PDOI 1, DLI 1) - Government college teacher vacancies filled (PDOI 2, DLI 2) - Improvement of teacher management system for nongovernment colleges (DLI 3)	Strategic planning and management capacity strengthened	Improved equity in access, quality, and relevance in college education system
Component 2: Improving Teaching and Learning Environment - Develop Institutional Development Plan (IDP) for 120 colleges. - Implement IDG program. - Upgrade basic teaching facilities. - Introduce institutional self-assessment. - Develop content and material for teacher trainings. Develop LMIS and e-content	- IDG implemented and teaching-learning facilities upgraded in 120 colleges. 110 colleges connected to BdREN and 94 with campus network. 35 colleges completed self-assessment for quality assurance (IRI 3) 8,306 teachers and managers in Honours and Masters colleges trained in subject, pedagogical, and managerial training (PDOI 3, DLI 5) - Online learning platform established (DLI 6)	- Improved teaching and learning environment in selected government and nongovernment colleges (PDOI 4, DLI 4) - Management in the participating colleges strengthened (IRI 2) - Increased satisfaction levels of students and teachers in grant supported colleges about teaching and learning environment (IRI 4) - College system's resilience built	Teaching and learning environment improved in participating colleges	



Notes: BdREN = Bangladesh Research and Education Network; DLI = Disbursement-linked indicator; IDG = Institutional Development Grant; IRI = Intermediate results indicator; LMIS = Learning Management Information System; NTRCA = Nongovernment Teachers' Registration and Certification Authority. Statements in italics are dropped activities; statements in bold italics are revised activities introduced through restructurings.

Project Development Objectives (PDOs)

6. The PDOs were to strengthen the strategic planning and management capacity of the college education subsector and to improve the teaching and learning environment of participating colleges.

Key Expected Outcomes and Outcome Indicators

7. The PDO has two objectives with the following outcome indicators:

- **Objective 1:** To strengthen the strategic planning and management capacity of the college education subsector. Measured by PDO Indicator 1: Sector strategy and action plan developed, adopted, and the initial implementation started and PDO Indicator 2: Government college teacher vacancies filled.
- **Objective 2:** To improve the teaching and learning environment of participating colleges. Measured by PDO Indicator 3: Number of teachers and managers in Honours and Masters colleges trained in subject and pedagogical training under the Project and PDO Indicator 4: Teaching and learning environment at Honours and Masters colleges upgraded.

Components

8. The CEDP consisted of three components (further details of the project components and subcomponents can be found in annex 8).

9. **Component 1: Strengthening Strategic Planning and Management Capacity** (estimated cost: US\$48 million; actual cost: US\$48 million). The objective was to address planning, governance, and management challenges and place reforms at the system level. The component consisted of two subcomponents: Subcomponent 1.1, which aimed to develop a strategic plan for the college subsector, linked to DLI 1, and Subcomponent 1.2, which aimed to improve the teacher management system, linked to DLIs 2 and 3.

10. **Component 2: Improving Teaching and Learning Environment in Participating Colleges** (estimated cost: US\$42 million; actual cost: US\$42 million). This component was to address the challenge of providing quality and relevant college education at the institutional level. The component consisted of two subcomponents: Subcomponent 2.1, which aimed to improve teaching-learning environment in colleges, linked to DLI 4, and Subcomponent 2.2, which aimed to strengthen teaching and management capacity in colleges, linked to DLI 5.

11. **Component 3: Project Management, Communication, and Monitoring and Evaluation** (estimated cost: US\$10 million; actual cost: US\$4.1 million). The objective was to support project management, communication, and monitoring and evaluation (M&E). It supported the establishment of a Project Management Unit (PMU), capacity building, and technical assistance necessary for project implementation and several analytical studies.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION (IF APPLICABLE)

12. The project was restructured four times, including one extension of the closing date and adjustments to DLIs and results indicators. Details of changes introduced under the four restructurings are presented in annex 7.



Revised PDOs and Outcome Targets

13. The PDOs were not revised. End targets of the indicators were revised, as described in paragraph 14.

Revised PDO Indicators

14. Three PDO indicators were revised as follows. Further details are provided in annex 7.

- The end-of-project (EOP) target for PDO Indicator 1 was revised in the second restructuring (June 2021) to replace ‘the first-year activities have been implemented’ with ‘MOE prepared an investment plan for quality enhancement of college education in line with the college subsector strategic plan’. The main rationale was (a) the original end target did not fully take into account the Government’s complete policy and procedures to develop the NSP, secure additional funding through approval of a DPP, and initiate activities based on an NSP, which would go beyond the project time frame; and (b) the Government’s change in policy, introduced in October 2016⁴ (after project approval), which made the development and approval of a Pre-Development Project Proposal (Pre-DPP), including a costing or investment plan, as mandatory before initiating the process for a DPP (see Annex 10 for detailed chart of approval steps).
- The description for PDO Indicator 3 was revised in the first restructuring (May 2019) to include ‘Certificate of Completion’, reflecting the addition of the enhanced continuous professional development (ECPD) course to Component 2 activities.
- The EOP target for PDO Indicator 3 was revised in the second restructuring, from 8,000 to 5,000 teachers and managers trained, due to travel restrictions and no in-person trainings during the COVID-19 pandemic and limited capacity of online training, which had just been introduced. The EOP target was increased in the third restructuring (May 2023) from 5,000 to 6,500 teachers and managers trained, as the capacity to deliver online training had increased by then, including through the availability of project-trained master trainers.
- The description for PDO Indicator 4 was revised in the first restructuring to remove the reference to a second round of the IDG process, reflecting the reduced number of IDG rounds from two to one (while maintaining the same number of grants to colleges).

15. These revisions did not affect project scope, impact, and achievement of objectives. For PDO Indicator 1, given the inapplicability of the original end target and change in the Government’s policy for initiating new development activities, the revision helped align the project’s activities to ground reality and with the Government’s new procedures. For PDO Indicator 3, the EOP target was reduced during the second restructuring and then increased in the third restructuring. The final number of teachers and managers trained exceeded the original EOP target. For PDO Indicator 4, the number of IDG rounds was reduced from two to one to ensure that the project implementation timeline was maintained without reducing the planned number of IDG grants to colleges.

Revised Components

16. **Component 1.** The project initially aimed to start college development activities aligned with the approved NSP. However, due to the abovementioned reasons for changing PDO 1, the activity was revised to developing an investment plan in line with the NSP. Additionally, under this component,

⁴ GoB. 2016. *Government Development Project Submission, Processing, Approval and Revision Method*. Planning Commission: Dhaka.



the NTSC was to be established to promote competitive and merit-based teacher recruitment. Instead, the MOE opted to achieve these aims by amending the NTRCA Act 2005.

17. **Component 2.** This component was revised to (a) reduce the number of IDG rounds from two to one and (b) revise teacher training design by adding practice-based ECPD courses for managers and teachers. Unutilized funds (US\$2 million) from Component 3 were reallocated to Component 2 to train an additional 1,500 teachers and managers, as the capacity of the NU system to deliver online training had increased. In addition, establishing an LMIS and developing e-content were added to enhance the sector's distance education capabilities. However, the development of e-content was dropped following delays in procuring technical assistance and the NU's decision to incorporate the activity under its broader program.

Other Changes

18. Changes were made to some IRIs to account for the 18-month extension and to DLIs and disbursement-linked results (DLRs) to align with changes made to Components 1 and 2 (details in annex 7).

Rationale for Changes and Their Implication on the Original Theory of Change

19. **Changes introduced through the restructurings were responses to implementation experience and external shocks.** The first restructuring revised the description of some indicators and extended the project closing date by 18 months, from June 30, 2022, to December 31, 2023, to enable achievement of the PDOs, which were adversely affected by slow implementation in the first two years due to challenges with establishment of the PMU and project management capacity. The second restructuring enabled the project to respond to the COVID-19 pandemic, which had led to the closure of all educational institutions for 18 months and placed travel restrictions, leading to the cessation of in-person trainings. The third restructuring responded to the Government's policy changes and enhanced training implementation capacity. Overall, the changes helped refine results indicators, expedite implementation, meet the evolving needs on the ground, and remove activities that were no longer relevant and reallocate funds to areas of greater need. These changes did not affect the original ToC.

II. OUTCOME

A. RELEVANCE OF PDOs

Assessment of Relevance of PDOs and Rating

20. The relevance of PDOs is rated High. At closing, the PDOs remained fully aligned with the World Bank's Country Partnership Framework (CPF, 2023–2027, Report No: 181003-BD) priorities. The PDOs contributed to the CPF High-Level Objective B on Improved Socioeconomic Inclusion, which aims to improve quality and equitable access for human capital development services. The project developed the NSP to establish directions and goals for the college sector and provide a consensus-based plan of action closely aligned with the National Education Policy 2010, Strategic Plan for Higher Education 2018–2030, Eight Five-Year Plan, and Government's Vision for 2021 and 2041. The project improved the quality of teaching and learning facilities in colleges through upgrading of classroom facilities, establishment of science and computer laboratories, access to high-speed internet connectivity, and provision of books and journals. The project addressed important quality concerns by reducing government college teacher vacancies and governance concerns through reforms to support merit and need-based teachers' recruitment in the nongovernment colleges. The participating colleges predominantly served disadvantaged groups, promoting equitable access to higher education.

**B. ACHIEVEMENT OF PDOs (EFFICACY)****Assessment of Achievement of Each Objective/Outcome**

21. Given that the project scope remained the same, the split ratings approach has not been applied as per the ICR guidance.

22. **Strengthening the strategic planning and management capacity of the college education subsector (Objective 1).** Achievement of this objective is rated Substantial, as all related results were achieved (table 2).

Table 2. Objective 1 - Summary of Indicators and Results Achieved

Indicators	Baseline	EOP Target	Actual Results	Achievement
PDO indicator 1: Sector strategy and action plan developed, and the initial implementation started	No strategic plan is available for the tertiary college subsector	The MOE prepared an investment plan for quality enhancement of college education in line with the college subsector strategic plan.	The MOE prepared an investment plan for quality enhancement of college education in line with the college sub-sector strategic plan.	Achieved
PDO indicator 2: Government college teacher vacancies filled	Vacancies: 2,700	The MOE has filled a cumulative total of 2,700 government college teacher vacancies. The MOE has completed a needs assessment for teacher posts.	The MOE filled 3,500 lecturer post vacancies in government colleges. The MOE has completed a needs assessment for teacher posts.	Overachieved
IRI 2: Management in the participating colleges strengthened (% of grant supported colleges publish timely annual reports) (percentage)	0	80	86	Overachieved
IRI 6: Midterm and follow-up tracer study conducted	0	Follow-up tracer study conducted	Follow-up tracer study conducted (Midterm tracer study completed in 2021)	Achieved

23. **The approval of the college sector strategy and action plan significantly strengthened the strategic planning capacity of the subsector.** The NSP for Higher Education Colleges in Bangladesh 2023–2031 was developed, approved, and published by the MOE (PDO Indicator 1). This is the first strategic plan for the tertiary college subsector. Its preparation was supported by six background studies on different aspects prepared by working groups whose expert members were from Bangladesh's public and private higher education sector and industry. The NSP provides an action plan to improve sector outcomes in the following areas: (a) vision, size, and shape; (b) access and equity; (c) quality and relevance; (d) management of the college system; (e) science, technology, and information and communication technology (ICT); and (f) financing. An investment plan based on the NSP was discussed in regional and national consultations and then approved by the Secondary and Higher Education Division (SHED) on August 6, 2023. Using the investment plan, the MOE prepared and submitted a Pre-DPP to the Economic Relations Division to initiate the process of allocating government and donor funds for continued development of the college subsector.

24. **The college subsector's teaching and teacher management capacity were strengthened to become more systematic and informed.** The teacher management system was significantly improved through the recruitment of 3,500 government teachers, far exceeding the target of 2,700 (PDO Indicator 2). In addition, a needs assessment for teacher posts in government colleges was prepared



by the Directorate of Secondary and Higher Education (DSHE) to inform the recruitment needs for the next three years. The project also supported a critical reform in the teacher recruitment procedures for nongovernment colleges by placing an amendment to the NTRCA Act 2005 to the Cabinet for approval, which specified criteria for competitive and merit-based teacher recruitment.

25. **The IDG program strengthened the management skills of participating colleges.** Of the 120 IDG colleges, 103 (86 percent) published annual reports for 2023 in a timely manner (IRI 2). Before the project, NU-affiliated colleges were not publishing annual reports regularly. The annual reports provide critical information on college activities, development projects, student and teacher information, academic programs, and financial information. These reports allow public access to information for better accountability and informed student choice.

26. **Sector planning capacity was enhanced by data collected from tracer studies of graduates** (IRI 6). The baseline (March 2019), first round (June 2021), and follow-up (October 2023) tracer studies provided a comprehensive picture and novel datasets on college graduate labor market outcomes and provided recommendations for improving employability of graduates. Their dissemination initiated national policy dialogue on poor graduate outcomes and informed the NSP's shift to a more outcome orientation, such as increased use of management information systems, internet connectivity, and campus networks which supports academic activities (annex 12 provides a summary of findings from the studies).

27. **The new LMIS was an important contribution to manage and deliver quality distance learning in the colleges and build resilience.** The LMIS is helping connect the NU with its extensive network of colleges across Bangladesh and providing online training and resources to teachers and students.

28. **Improving the learning and teaching environment of participating colleges (Objective 2).** Achievement of this objective is rated Substantial. Four of the five related targets were achieved, with one partially achieved (table 3).

Table 3. Objective 2 - Summary of Indicators and Results Achieved

Indicators	Baseline	EOP Target	Actual Results	Achievement
PDO indicator 3: Number of teachers and managers in Honours and Masters colleges trained in subject and pedagogical training under the Project	0	6,500 college teachers and managers are trained	8,306 teachers and managers trained	Overachieved
PDO indicator 4: Teaching and learning environment at Honours and Masters colleges upgraded (% of IDP activity milestones achieved)	0	70	84	Overachieved
IRI 1: Students benefiting from direct interventions to enhance learning (percentage of whom are female)	0	2,100,000 (50 percent female)	2,342,802 (48 percent female)	Overachieved (female target, mostly achieved)
IRI 3: Number of colleges which completed self-assessments for quality assurance	0	30	35	Overachieved
IRI 4: Increased satisfaction levels of students and teachers of grant supported colleges about teaching and learning environment	Student: 2.57 Teacher: 2.42	Student: Teacher: (30 percent increase from the baseline)	Student: 3.53 (37.3 percent increase from baseline) Teacher: 2.68 (10.7 increase)	Partially Achieved



Indicators	Baseline	EOP Target	Actual Results	Achievement
			from the baseline)	

29. **Teachers and managers were trained to international standards.** The teaching quality and management capacity of participating colleges were systematically and sustainably improved through (a) training of master trainers, (b) teacher training, (c) capacity building for future college leaders and principals, and (d) capacity building for policy makers and subsector leaders. At project closing, 8,306 teachers and managers had been trained (details are in annex 11), exceeding the revised EOP target of PDO Indicator 3 (6,500 teacher and managers) and the original EOP target of 8,000 teachers and managers trained. The project promoted continuous professional development by establishing a Training Consortium, comprising four national agencies (NU, Directorate of Secondary and Higher Education (DSHE), Bangladesh Open University (BOU), and National Academy of Education Management (NAEM)) and the University of Nottingham Malaysia (UNM), as an international academic partner, to ensure global best practice. The teacher trainings were undertaken based on a teacher training needs assessment conducted by UNM and revised based on stakeholder feedback during implementation. An evaluation of the teacher training program confirmed positive outcomes, including improved pedagogy/andragogy skills, subject knowledge, and overall teaching quality.⁵ Notably, the local pool of master trainers will enable continuation of the teacher professional development program.

30. **Teaching and learning facilities were upgraded.** The IDG program was successfully implemented, with 120 grantee colleges achieving 84 percent of their IDP milestones (PDO Indicator 4) and utilizing an average of 86 percent of allocated grants. The IDGs were used to refurbish 1,889 classrooms and establish 158 ICT labs, 269 science labs, 624 libraries, and 62 mothers' corners. The BdREN connection was extended to 110 colleges, and campus networking was established in 94 colleges, enhancing access to high-speed internet and global academic resources. An evaluation shows that the IDG program has significantly improved the teaching and learning environment, particularly in poorly equipped colleges.⁶ A summary of IDG activities and results is in annex 9. The management capacity of over 9,988⁷ teachers and managers was strengthened in areas such as procurement, financial management, ICT, office management, recordkeeping, and M&E.

31. **Institutional self-assessments helped colleges in reviewing, assessing, and developing action plans to improve their teaching-learning environment and quality.** As part of the IDG program, 35 grantee colleges completed institutional self-assessments (IRI 3), a first in the sector. The self-assessments were used to review the quality of teaching, curriculum, and program delivery and gathered feedback from teachers and students on the teaching-learning environment in colleges. After an external peer review of self-assessments, the colleges prepared IDPs to guide continuous quality assurance activities.

32. **At project closing, more than 2.34 million students (48 percent female) benefitted from project interventions (IRI 1).** Three rounds of surveys were conducted to understand students', teachers', and principals' satisfaction with, and employers' satisfaction with students from, grantee colleges. At project closing, students' satisfaction level increased by 37.3 percent while teachers' satisfaction level increased by 10 percent, compared to the baseline. Improvements in learning environment and teaching quality motivated students to engage more in academic and labs activities and improve class attendance. Several grantee colleges signed partnership memoranda of

⁵ Samahar Consultants Ltd. 2022. *Report on Effectiveness and Situation Assessment of Teacher's Training*. CEDP, National University, MOE.

⁶ BIDS. 2023. *Study on Project Effectiveness including Endline Satisfaction Survey of College Education Development Project (CEDP)*. National University, MOE.

⁷ These are teachers and managers who were given additional trainings on grant/project management and not included in the number achieved in PDO Indicator 3, which focused on subject-based and pedagogical training.



understanding with industries and research organizations, initiated activities to promote soft skills of students, and arranged job fairs in college campuses. Teachers reported increased satisfaction with the teaching-learning environment, quality of academic infrastructure, and access to ICT facilities, but lower overall satisfaction related to the quality of students' soft skill development and opportunities to collaborate with industry. While the NU has taken steps to introduce new soft skills development programs in 2022, there is a clear need for more investment in such programs (results from the surveys are summarized in annex 12).

Justification of Overall Efficacy Rating

33. Overall efficacy is rated Substantial based on the achievement of project objectives as described above. The project exceeded or almost fully achieved its objectives.

C. EFFICIENCY

Assessment of Efficiency and Rating

34. **Economic analysis.** The economic analysis at appraisal used collective costs and aggregated individual benefits to estimate the project's prospective returns. Benefits include (a) increased employability of graduates and (b) increased likelihood of students completing the college program. Costs include the project costs and the opportunity costs in foregone income (using the annual wage of Grade 12 graduates) and out-of-pocket expenditures of dropped-out students for two years. In the mid to long run, the CEDP was expected to create significant positive impact on the society by (a) creating a more productive and adaptable labor force for various economic sectors and public agencies, (b) producing more capable teachers for public and private schools and improving the quality of primary and secondary education, and (c) contributing to more favorable investment environments to attract domestic and foreign companies by providing larger pools of capable recruits. This ICR conducted a cost-benefit analysis using the same approach as during project preparation. The ex-post analysis found that the present value of net benefits is positive (US\$620 million) and the project's IRI is 14 percent, which is higher than the discount rate (see annex 4 for details).

35. **Efficiency in design and implementation aspects.** The project design utilized local and in-house expertise, in collaboration with UNM (an international partner), to enhance the institutional capacity of the college sector in delivering cost-effective academic and professional development training. Notably, teacher training involved key partnerships with four government entities—NU, BOU, DSHE, and NAEM—leveraging their facilities and logistical support, significantly reducing costs compared to full outsourcing. The training program adopted a cascade approach, with a cohort of college teachers completing the Master's in Education program at UNM. These teachers then served as master trainers, delivering professional development training in Bangladesh under the quality supervision of UNM. The CEDP developed a Project Management Information System (PMIS), which helped collect IDG-related data from 120 colleges spread all over Bangladesh in real time and reduced the cost of manual data collection, entry, and analysis. However, the project required a no-cost 18-month extension to make up for slow implementation in the first two years. The efficiency of project implementation was demonstrated during the pandemic when, despite the 18-month closures in the college sector, no further extensions were needed, and the project was able to close successfully. The project also continued IDG activities through e-procurement and online teacher training during the pandemic.

36. At closure, the project utilized 94 percent of total committed funds, and the remaining unused 6 percent was cancelled during the fourth restructuring. The cancellation included savings from the technical assistance component; planned small works and services that were not required; and the e-contents activity, which was decided to be pursued under the Government's broader development program.



37. The project Efficiency is rated Substantial based on high economic returns, cost-efficient project design, and substantial implementation efficiency. The efficiency is what would be expected in the higher education sector.

D. JUSTIFICATION OF OVERALL OUTCOME RATING

38. The overall outcome rating is Satisfactory given High Relevance, Substantial Efficacy, and Substantial Efficiency, and that the efficacy of both objectives was rated Substantial.

E. OTHER OUTCOMES AND IMPACTS (IF ANY)

Gender

39. The project tracked gender aspects using an indicator: ‘Number of beneficiaries (NU-affiliated tertiary level college students)’, revealing 48 percent female beneficiaries among around 2.3 million total participants in the college sector (slightly below the 50 percent target). The IDG program was designed with a dedicated window to support 31 girls’ colleges (out of 300 girls’ college nationwide). The IDGs allowed girls’ and coeducational colleges to (a) upgrade teaching and learning facilities, including science and computer labs to enable more female students to pursue higher education in science; (b) improve sanitation and safety facilities to create female-friendly environments in the colleges; and (c) improve soft skills and industry links to enhance female college graduate employability.⁸ Furthermore, 62 mothers’ and childcare corners were established in women’s and coeducational colleges to meet female students’ and teachers’ needs. The project also provided educational leadership and subject-based training to 1,567 female teachers in the college sector. The sector saw a 50 percent increase in female enrollments in colleges as well as a 7-percentage point increase in female gross enrollment rate in tertiary education during the project implementation period (see annex 13 for data on the accelerated enrollment during the project period).

Institutional Strengthening

40. Institutional strengthening was embedded in the project’s objective. The project supported strategic interventions to improve system- and institutional-level capacity building, a first in the college subsector. At the national level, the development of the strategic plan and teacher recruitment reforms improved the planning and management capacity of the sector. At the college level, the recruitment of additional teachers reduced a critical institutional-level gap in delivering good quality education. The IDG program helped colleges improve their planning, management, and academic capabilities through institutional self-assessments, training of managers and teachers, and connectivity to global resources. The project-supported training helped create a pool of master trainers to provide subject-based and pedagogical training for the college subsector. The Government has extended the BdREN network to an additional 110 colleges.

Poverty Reduction and Shared Prosperity

41. The project invested in colleges in the urban periphery and rural areas, bringing access to better quality higher education to students from socioeconomically disadvantaged families. The IDG activity was designed based on geographic location and enrollment size to ensure equity in grant award to colleges from disadvantaged areas. Around 50 percent of IDGs were awarded to colleges located in remote and rural areas. Access to quality facilities, including science and computer labs, and teacher training in these colleges were critical to open more opportunities for students and facilitate future poverty reduction for poor families and became extremely important during the pandemic.

⁸ BIDS (Bangladesh Institute of Development Studies). 2023. *Study on Project Effectiveness Including Endline Satisfaction Survey*. Dhaka: Bangladesh.



III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

42. **Well-aligned project objectives and well-informed choice of interventions.** The choice of project objectives was supported by activities at the system and institutional level identified as critical to achieve the sector's mid- to long-term development. The project tackled the core issues to build the college system's capacity to plan, manage, and implement institutional programs and bring reforms to strengthen the foundation for continued development. Given the constraints of limited resources, the project design appropriately prioritized key systematic reforms, maintaining simplicity for successful implementation. At the institutional level, interventions were focused on 120 government and nongovernment colleges, a subset of the over 1,700 colleges in Bangladesh, specifically targeting Honours and Master Programs. The project design also focused on the quality of teaching, including teacher recruitment and professional development, which are considered foundational elements for college development.

43. **Lessons learned from previous operations in higher education.** The project built on successful activities and incorporated lessons learned from the earlier World Bank operation for universities, the Higher Education Quality Enhancement Project (HEQEP, P161386). The expansion of BdREN connectivity, introduction of quality assurance in college sector, and the use of competitive IDG were included in the CEDP based on the experiences of HEQEP.

44. **Adequate risk assessment.** Most of the project risks were adequately assessed and addressed. Based on the identified risks, mitigation measures were developed, with strong focus on continuous support and training to the PMU staff and IDG-supported colleges. However, the risk assessment for institutional capacity to implement fell short, as reflected by the initial implementation challenges, given that the project was a first-time intervention in the sub-sector. It was not reasonable to expect the external shock of the pandemic.

B. KEY FACTORS DURING IMPLEMENTATION

Factors Subject to the Control of Government and Implementing Entities

45. **Dedicated involvement of education stakeholders improved institutional capacity.** The project actively engaged college teachers, education experts, and administrators from the NU, NAEM, BOU, and DSHE in the planning, development, and implementation of key project activities. Development of the NSP was led by in-country education experts and teachers, which ensured ownership of the deliverables. Various trainings were provided to college teachers and education administrators across 120 colleges on grant planning, implementation, monitoring, and fiduciary management, which significantly contributed to autonomy and decentralized management in the colleges.

46. **Development of a Training Consortium with national and international partners helped increase and sustain training capacity in the college subsector.** At the preparation stage, the NU had capacity to train only about 1,200 teachers annually against the more than 60,000 teachers in the system. The delivery of training to college teachers also relied heavily on the availability of university teachers as trainers. The creation of a Training Consortium comprising four national agencies and UNM as the international academic partner helped increase training capacity, bringing more resources (training venues, facilities, and trainers) and inculcating global best practices in teacher professional development programs. Throughout implementation, the project engaged with these agencies and held them accountable for results by requiring submission of periodic progress reports and including them in supervision missions.



47. **There were staffing challenges.** The project experienced delays during the first two years of implementation. The delay in the deployment of a project director and key technical specialists, including procurement and financial management specialists, as well as the frequent turnover of project leaders (five times in two years) led to slow decision-making. Following revamped project leadership, an effective MTR, and continuous hands-on support for capacity building, the pace of implementation picked up and the project performed well until closure.

48. **Lengthy government approval process challenged the timely implementation of project activities.** The project's implementation progress faced challenges due to the prolonged approval processes by the Government. The approval of the first restructuring took 12 months, while the second approval took almost 21 months. A contributing factor to the delay was the necessity to establish consensus among the various partner agencies. The delay in the second approval delayed the release of performance grants to colleges, which could only be achieved in the last year of project implementation.

Factors Subject to World Bank Control

49. **Effective restructurings were undertaken to address implementation challenges and external shocks.** Following the MTR in January 2019 and subsequent first restructuring, ratings improved, and the project was removed from problem status. The World Bank team undertook three more restructurings to adapt to policy changes, integrate stakeholder feedback, and mitigate pandemic impacts. Training programs were tailored to evolving needs and participants' feedback, while DLRs were adjusted to align with the Government's decision on the new approach to nongovernment teachers' recruitment reform. Additionally, online training initiatives and the development of an LMIS were introduced to support distance learning in response to the pandemic.

50. **Capacity building was provided by the World Bank team throughout project implementation.** Project implementation was intensively monitored, and continuous support was provided to the PMU and partner agencies as changes were introduced through restructurings. Given the lack of experience of the college teachers and staff in project implementation, the continuous capacity building and close on-the-ground support helped address issues quickly.

51. **The initiative to use electronic government procurement (e-GP) helped speed up project procurement.** The application of e-GP in colleges was a key factor, which expedited implementation by significantly reducing procurement time, increasing transparency, and allowing real-time monitoring.

Factors outside the Control of Government and Implementing Entities

52. **Restrictions during the COVID-19 pandemic hampered project activities.** The COVID-19 outbreak first occurred in Bangladesh in March 2020, with all education institutions, including colleges, remaining closed for 18 months. Government offices were periodically shut down or operating with restrictions, which often slowed the decision-making and approval process. In addition, many key government officials contracted the virus during the project period, requiring more time for approvals and processing of activities.

53. **Restrictions on foreign training adversely affected the in-person training results.** Since 2022, the Government set restrictions on foreign training in response to macroeconomic fiscal pressures on foreign reserves. This interrupted the in-person professional development trainings at the UNM campus in Malaysia. In response, the last cohort of the master trainer training was conducted virtually to meet project targets.



IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M&E Design

54. **The project Results Framework was adequately designed with minor shortcomings.** The project objective was clearly defined and well aligned with the Government's strategy and tracked by meaningful PDO-level and intermediate-level indicators. The PDO indicators and intermediate-level outcomes were clear, easy to understand, measurable, and relevant. The M&E included semiannual progress reports and studies on graduate tracking surveys, teacher training effectiveness, and satisfaction surveys to monitor implementation. The DLIs were linked to the Results Framework, and the verification protocol and methodology were clearly defined. The design of the results indicators, however, had some shortcomings. In the case of PDO Indicator 1, the end target was unrealistically set up and had to be revised to what was achievable within the project's time frame and in line with the original indicator. It also contributed to strengthening the basis for implementation of the NSP beyond the project period.

M&E Implementation

55. **The project M&E was well implemented.** The Results Framework and DLIs were closely monitored and timely adjusted throughout project implementation to reflect changes. DLI verification protocols were complied with and reported on adequately. The PMU maintained and submitted detailed semiannual monitoring reports in conjunction with the World Bank's biannual implementation support missions in a timely manner. By project closure, 14 high-quality semiannual reports had been submitted to the World Bank. College-level annual reports were also prepared and published by 86 colleges on their websites. The PMIS was established and was fully functional, facilitating the consolidation and reporting of data. In addition, several high-quality studies and surveys to inform project and sector interventions were undertaken. Six background research papers were prepared for the development of the NSP. Moreover, the project completed two rounds of graduate tracer studies; baseline, midterm, and endline satisfaction surveys; a teacher training evaluation study; and a project effectiveness study.

M&E Utilization

56. **M&E data were actively utilized throughout the implementation of the project.** The semiannual progress reports helped facilitate mission dialogue and make the missions more effective, as the team had time to utilize the information to plan before each mission. The results and recommendations from the studies and surveys were used to improve project efficacy and efficiency and continue to support policy dialogues among education stakeholders.

Justification of Overall Rating of Quality of M&E

57. **The overall quality of M&E is rated Substantial.** M&E was well designed, monitored, and utilized. The project achievements and lessons learned were captured well in the government reports. Results from studies and surveys effectively supported project management. The moderate shortcomings/misalignment of any indicators were proactively solved through project restructurings.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

Environmental and Social Safeguards

58. At project closure, the adherence to environment and social safeguards was satisfactory. Full compliance with safeguards' due diligence, grievances management, occupational health and safety, code of conduct, and gender-based violence prevention training was ensured. Regular consultations with stakeholders and timely submission of progress reports were maintained. The project triggered



OP/BP 4.10 for social safeguards but posed no adverse impact on small ethnic and vulnerable communities (SEVC). An SEVC Development Framework and Gender Action Plan were implemented. Minor worker injuries were promptly handled. Ten grievances related to personal protective equipment were recorded and satisfactorily addressed. The project addressed some challenges in implementing environment and social safeguards such as staff turnover and small occupational health and safety budget. Necessary safeguards training for the PMU and contractors further improved their institutional capacity to monitor and manage environment and social aspects of the project activities.

Procurement

59. The overall procurement management performance was satisfactory. Despite the initial delays, all procurement activities were completed. Measures taken to strengthen the procurement capacity included engagement of a full-time procurement consultant, training of project officials on procurement, and targeted implementation support by the task team. Significant progress was made by June 2022, when most of the contracts were awarded. Overall, 135 packages were procured by the PMU, including 9 high-value packages for IDG; 1,624 packages were procured by colleges, including 891 packages that were procured through e-GP; and 100 networking packages were awarded. Capacity was developed through e-GP training to 236 participants and procurement training to 143 participants.

Financial Management

60. The overall financial management performance of the project was satisfactory. All financial reports, including the interim unaudited financial reports, were submitted on time with acceptable quality. Financial management training using the PMIS was provided to all 120 colleges. Internal audits introduced under the project were adequately completed and two internal audit reports were submitted to the World Bank. The IDG grantee colleges maintained accounting books and updated the accounting records in the system in a timely manner.

C. BANK PERFORMANCE

Quality at Entry

61. The project design was highly relevant and aligned well with the government strategy to improve the quality of and access to higher education. Project interventions were well targeted to the challenges of the college sector. Most risks were adequately identified, with a mitigation plan that focused on continuous capacity building and hands-on training. However, risks related to delays in establishment of the PMU and subsequent implementation capacity issues in the initial years had not been expected to be as serious as they turned out to be. The M&E was designed to include studies to support implementation and assess project performance, although PDO Indicator 1 had some shortcomings.

Quality of Supervision

62. The project was closely supervised and effectively supported by the World Bank team throughout implementation. In total, 12 formal implementation support missions were conducted in addition to weekly and monthly meetings with the NU, PMU, and grantee colleges. Frequent and pertinent discussions were held with the Government to find solutions for implementation challenges. With the staffing enhancement for the PMU, the project was successfully turned around from problem status in 16 months following an effective MTR. The team proactively carried out restructurings based on implementation experience and to respond to external shocks. The first restructuring took time to initiate, mainly due to the need for project leadership and the PMU to be in place and enable the World Bank team to conduct effective restructuring discussions. The Aide Memoires were detailed and candid with descriptions of the progress, challenges, and recommendations across each activity. Time-bound action plans for the next six-month period were included in the Aide Memoires.



Proactiveness in cancellation of unused funds and extension of the closing date allowed the project to close successfully.

Justification of Overall Rating of Bank Performance

63. The overall World Bank performance is rated Satisfactory. The project was adequately designed and appraised. The implementation of the project was closely monitored and supported with timely corrective measures.

D. RISK TO DEVELOPMENT OUTCOME

64. The risk to development outcome is Low. The Government demonstrated a strong commitment to the reform by approving the NSP and investment plan, laying a robust foundation for the college subsector's development. Capacity was significantly enhanced at all levels, from policy makers to college principals and teachers, through training by the UNM, NU, PMU, and World Bank, as well as through project implementation experience. Funds for operation and maintenance of facilities financed by the project are maintained through the annual college budgets. Results of the IDG program were highly evaluated by the Government and scaled up under the Government's programs after project closure.

V. LESSONS AND RECOMMENDATIONS

65. **Simple and prudent design for first time sector operations is important.** The CEDP focused on selected interventions for system and institutional capacity building. The project demonstrated realism in acknowledging the time required for government consultations and approval processes of the NSP and the teacher management reform agenda, which can span several years due to the political and complex nature of system-level reforms.

66. **The use of results-based financing was effective for achievement of education reforms.** The use of DLIs provided strong motivation to the Government in pursuing challenging reforms in areas of sector planning and teacher management. The DLIs helped build consensus among different stakeholders on expected results and achieve the results.

67. **Timely establishment of project management is critical for building and sustaining implementation momentum and ownership of project.** The first two years' delay in establishment of the PMU and subsequent implementation challenges highlight the need for greater dialogue with the Government to ensure project implementation readiness before approval. This could mean, for example, assigning a project leader from the Government before approval. The CEDP's later implementation success was due to strong commitment and full engagement from the Government, which enabled quick staffing of the PMU after initial delays, timely measures to enhance staff capacity, and maintenance of a strong team until closure.

68. **For novel activities, continuous capacity building is key to project success.** Originally, the CEDP planned the IDG implementation cycle to be completed over three years. However, due to a combination of low implementation capacity and 18 months' closure of college caused by the COVID-19 pandemic, actual completion of the IDGs took nearly four years. The capacity at colleges to plan, implement, monitor, and report on IDGs was developed through continuous formal and hands-on trainings throughout the life of the project.

69. **Leveraging partnerships is an effective approach to building and sustaining training capacity.** The establishment of the Training Consortium proved to be highly effective. The trainings delivered under this arrangement were not only of high quality but also a cost-effective means to enhance capacity and sustain teacher professional development in the college subsector. The consortium arrangement allowed sharing of training venues and resources to deliver training to more teachers.



Master trainers are now able to effectively lead and deliver training as NU resource persons, reducing dependence on costly external trainers.

70. **Ability to pivot and adapt to changing situations is critical to ensure project relevance.** The CEDP successfully completed all activities despite all the challenges. This can be credited to the project team's ability to adapt to changing situations, from both the client and the World Bank Group side. Throughout the pandemic and the foreign training embargo, trainings were continued by shifting to online mode and IDG procurement continued with remote support from the PMU and with stronger focus on e-procurement. On the World Bank Group side, restructurings to adapt to the changes and revisions to design supported the project in responding to the changing needs on the ground and remain relevant.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: To strengthen the strategic planning and management capacity of the college education subsector

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Sector strategy and action plan developed, and the initial implementation started	Text	No strategic plan is available for the tertiary college subsector.	The first year activities have been implemented	MOE prepared an investment plan for quality enhancement of college education in line with the college sub-sector strategic plan	MOE prepared an investment plan for quality enhancement of college education in line with the college sub-sector strategic plan
		03-Oct-2016	31-Dec-2021	29-Dec-2023	07-Nov-2023

Comments (achievements against targets):

The end target is fully achieved. The Strategic Plan for Higher Education Colleges 2023 - 2031 was approved on October 9, 2023 and published online. An investment plan for the quality enhancement of the college subsector based on the Plan was prepared and approved by SHED/MoE.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
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Government college teacher vacancies filled	Text	Vacancies: 2,700	MoE has completed a needs assessment for teacher posts	MoE has completed a needs assessment for teacher posts	MoE has completed a needs assessment for teacher posts
		03-Oct-2016	31-Dec-2021	29-Dec-2023	02-May-2022
Comments (achievements against targets): The end target is fully achieved.					
Objective/Outcome: To improve the teaching and learning environment of participating colleges					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of teachers and managers in Honours and Masters colleges trained in subject and pedagogical training under the Project	Text	0.00	8,000 college teachers and managers are trained	6,500 college teachers and managers are trained	8,306 teachers and managers trained
		03-Oct-2016	31-Dec-2021	29-Dec-2023	31-Dec-2023
Comments (achievements against targets): The end target was surpassed.					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion



Teaching and learning environment at Honours and Masters colleges upgraded (% of IDP activity milestones achieved)	Number	0.00 03-Oct-2016	70.00 31-Dec-2021	70.00 29-Dec-2023	84.00 07-Nov-2023
Comments (achievements against targets): The end target was surpassed.					

A.2 Intermediate Results Indicators

Component: Component 1: Strengthening the strategic planning and management capacity

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Students benefiting from direct interventions to enhance learning	Number	0.00 03-Oct-2016	2,000,000.00 31-Dec-2021	2,100,000.00 29-Dec-2023	2,342,802.00 31-Dec-2023
Students benefiting from direct interventions to enhance learning - Female	Number	0.00	1,000,000.00		1,124,924.00
Comments (achievements against targets): The end target was surpassed for total beneficiaries. Based on updated BANBEIS 2022 data:					
Total Female					



Degree 726,744 352,026
 Pass
 Honour's 1,425,207 687,591
 Masters 190,851 85,307
 Total 2,342,802 1,124,924

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Management in the participating colleges strengthened (% of grant supported colleges publish timely annual reports)	Percentage	0.00 03-Oct-2016	80.00 31-Dec-2021	80.00 29-Dec-2023	86.00 07-Nov-2023

Comments (achievements against targets):

The end target was fully achieved. A total of 103 colleges (out of 120) published annual reports for 2023.

Component: Component 2: Improving the teaching and learning environment in participating colleges

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of colleges which completed self-assessment for quality assurance	Number	0.00 03-Oct-2016	30.00 31-Dec-2021	30.00 29-Dec-2023	35.00 31-Dec-2023

Comments (achievements against targets):



The end target was surpassed. 35 colleges have completed the Institutional Self-Assessment (ISA) and Post Self-Assessment Improvement Plan (PSA IP).

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Increased satisfaction levels of students and teachers of grant supported colleges about teaching and learning environment (targets set by % increase)	Text	n.a	30% increase from baseline	30% increase from baseline	Student: 37.3% increase Teacher: 10.7% increase
		03-Oct-2016	31-Dec-2021	29-Dec-2023	31-Dec-2023

Comments (achievements against targets):

The end target was achieved for students; and partially met for teachers.

Component: Component 3: Project Management, communication and monitoring and evaluation

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
PMU is fully operational	Yes/No	No	Yes	Yes	Yes
		03-Oct-2016	31-Dec-2021	29-Dec-2023	07-Nov-2023

Comments (achievements against targets):



The indicator is fully achieved. Being fully operational is defined as having met three of the following four conditions: (a) Staffing (70 percent of PMU staff are in place); (b) Procurement (at least 70 percent of the procurement was completed as intended in the Procurement Plan); (c) M&E (at least 70 percent of the intended reports published and activities completed); and (d) FM (the previous IUFR is completed in time).

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Midterm and follow-up tracer study conducted	Text	n.a	Follow-up Tracer study conducted	Follow-up Tracer study conducted	Follow-up Tracer study conducted
		03-Oct-2016	31-Dec-2021	29-Dec-2023	07-Nov-2023

Comments (achievements against targets):

The indicator is fully achieved.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Web-based Project Management Information System (PMIS) is established and regularly updated.	Text	n.a	PMIS fully functional and regularly updated	PMIS fully functional and regularly updated	PMIS fully functional and regularly updated
		03-Oct-2016	31-Dec-2021	29-Dec-2023	07-Nov-2023

Comments (achievements against targets):

The indicator is fully achieved.



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Awareness raising activities are regularly conducted.	Number	0.00 03-Oct-2016	10.00 31-Dec-2021	16.00 29-Dec-2023	124.00 07-Nov-2023
Comments (achievements against targets): The end target was surpassed.					



B. KEY OUTPUTS BY COMPONENT

Objective/Outcome 1	
Outcome Indicators	1. Sector strategy and action plan developed, and the initial implementation started. 2. Government college teacher vacancies filled
Intermediate Results Indicators	1. Web-based Project Management Information System (PMIS) is established and regularly updated 2. Awareness raising activities are regularly conducted 3. Midterm and follow-up tracer study conducted
Key Outputs by Component (Linked to the achievement of the Objective/Outcome 1)	1. National Strategic Plan approved and published 2. Investment Plan approved 3. 3,500 government college teachers recruited 4. NTRCA Act 2005 amended 5. Tracer study conducted 6. Follow-up tracer study conducted 7. PMIS developed and fully functioned
Objective/Outcome 2	
Outcome Indicators	1. Number of teachers and managers in Honours and Masters colleges trained in subject and pedagogical training under the Project 2. Teaching and learning environment at Honours and Masters colleges upgraded (% of IDP activity milestones achieved)
Intermediate Results Indicators	1. Students benefiting from direct interventions to enhance learning (disaggregated by gender) 2. Management in the participating colleges strengthened (% of grant supported colleges publish timely annual reports) 3. Management in the participating colleges strengthened (% of grant supported colleges publish timely annual reports)



	4. Increased satisfaction levels of students and teachers of grant supported colleges about teaching and learning environment
Key Outputs by Component (Linked to the achievement of the Objective/Outcome 2)	1. 8,306 college teachers and managers trained in pedagogical training 2. 7,500 teachers trained in basic ICT, Operations Manual & Procedures, Procurement, Record keeping 3. 124 workshops organized including 5 national, 50 regional and 50 face-to-face and online self-assessment workshops, 11 workshops on IDG implementations 4. Student and Teacher Satisfaction surveys conducted at baseline, midterm and endline 5. Effectiveness and Situation Assessment of Teacher Training conducted 6. 94 colleges have internet connectivity and campus networking established 7. 35 colleges completed ISA and improvement plan 8. 2.34 million students benefited from the project, 48 percent of which are female 9. 1,889 classrooms, 286 laboratories, 158 ICT labs, 624 libraries, 534 seminar libraries, 44 auditoriums, 362 office rooms, 72 teacher councils, 299 teacher's rooms, 1,066 washrooms, 62 mothers' corners, 63 student common rooms, 21 language labs, 12 Muktijoddha/Bongobondhu corner renovated/ refurbished 10. 20,000+ lab equipment, 40,106 electronic and electrical equipment, 938,653 furniture and fixtures installed 11. Power stations and water treatment systems installed

**ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION****A. TASK TEAM MEMBERS**

Name	Role
Preparation	
Yoko Nagashima, Mokhlesur Rahman	Task Team Leader(s)
Ishtiak Siddique	Procurement Specialist(s)
Mohammed Atikuzzaman	Financial Management Specialist
Nazma Sultana	Team Member
Halil Dundar	Team Member
Hena G. Mukherjee	Team Member
Sandra X. Alborta	Team Member
Shakil Ahmed Ferdausi	Environmental Specialist
Tobias Linden	Peer Reviewer
Dhushyanth Raju	Peer Reviewer
Atou Seck	Peer Reviewer
Hisham A. Abdo Kahin	Team Member
Burhanuddin Ahmed	Team Member
Karthika Radhakrishnan	Team Member
Muhammad Asahabur Rahman	Team Member
Sabah Moyeen	Social Specialist
Francisco Marmolejo	Peer Reviewer
Ferdous Jahan	Social Specialist
Tashmina Rahman	Team Member



Rex Joseph Quiah	Team Member
T. M. Asaduzzaman	Team Member
Tanvir Ahmed	Environmental Specialist
Shiro Nakata	Team Member
Supervision/ICR	
Tashmina Rahman	Task Team Leader(s)
Arafat Istiaque	Procurement Specialist(s)
Md. Didarul Islam	Financial Management Specialist
Mohammed Atikuzzaman	Financial Management Specialist
Nusrat Jahan	Team Member
Md Zulfiquar Rahman	Team Member
Umme Saima Sadia	Team Member
Sayed Mujtaba Shobair	Environmental Specialist
Shourov Kumar Sharma	Procurement Team
Syud Amer Ahmed	Team Member
Sabah Moyeen	Social Specialist
Muhammad Asahabur Rahman	Team Member
Dung Thi Thuy Dao	Team Member
Sangeeta Goyal	Team Member
Sandra X. Alborta	Team Member

B. STAFF TIME AND COST

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US\$ (including travel and consultant costs)
Preparation		
FY15	11.130	87,300.23
FY16	35.003	232,528.29
FY17	.250	92,656.32



Total	46.38	412,484.84
Supervision/ICR		
FY17	22.330	156,673.39
FY18	26.086	154,036.06
FY19	20.857	120,784.27
FY20	40.516	175,652.48
FY21	29.520	149,423.24
FY22	28.141	144,433.96
FY23	26.322	128,061.03
FY24	23.064	119,976.66
Total	216.84	1,149,041.09

**ANNEX 3. PROJECT COST BY COMPONENT**

Components	Amount at Approval (US\$, millions)	Actual at Project Closing (US\$, millions)	Percentage of Approval
Component 1: Strengthening Strategic Planning and Management Capacity	48.0	48.00	100.0
Component 2: Improving Teaching and Learning Environment in Participating Colleges	42.0	42.00	100.0
Component 3: Project Management, Communication and Monitoring and Evaluation	10.0	4.12	41.2
Total	100.00	94.12⁹	94.0

⁹ US\$6.34 million (SDR 4.824 million) was cancelled in December 2023.



ANNEX 4. EFFICIENCY ANALYSIS

1. The CEDP supported a range of institution-level and sector-wide interventions to improve the quality and relevance of college education and the management and governance of the college subsector in Bangladesh. The potential benefit from the project is therefore likely to go well beyond individual beneficiaries and cut across different sectors. However, for the benefit of focusing the analysis, this cost-benefit analysis deliberates only the collective costs and benefits at the individual beneficiary level. The following thus presents an ex post cost-benefit analysis (in the form of improved graduation, employment, and earnings prospects) and the resulting internal rate of return (IRR) (the interest rate at which net present values [NPVs] of both costs and benefits are equal to zero) of the project. Sensitivity analysis findings—changes in the underlying parameters (such as potential future employment fluctuations and varying wage premiums from the quality enhancement of education because of the project) that affect anticipated outcomes—are also presented.
2. To the extent possible, the methodology outlined below explains how the ex post estimation of benefits and costs accrued from the CEDP should be viewed against a ‘counterfactual’ where the GoB would continue without the project.
3. Similar to the ex ante analysis, the analysis assumes that (a) students enrolled in the last three years of the project will receive more or less full benefits of the project and are regarded as potential beneficiaries and (b) college students graduate at the age of 22 and continue working until the age of 60. Table 4.1 summarizes the data used and the full set of assumptions made in the ex post analysis to calculate the project’s IRR and NPV. In cases where the assumptions are different from ex ante analyses, they have been explicitly outlined in table 4.1.

Table 4.1. List of Data and Assumptions for the Cost-Benefit Analysis

Item	Data/Assumption	Source	Difference with Ex Ante Analysis
Total project cost	US\$93,580,000	Project report	Updated
Total number of beneficiaries	2,342,802	Project report - achieved number at completion	Updated
% of coverage by CEDP	20	Project design	Same as before
Graduation rate before the project (%)	75	NU	Same as before
Graduation rate after the project (%)	78.07	Tracer study data	Updated
National tertiary employment rate without CEDP (%)	88.8	Labour Force Survey	Updated
National tertiary employment rate with CEDP by scenario (%)	88.8, 89.8, 90.8	Sensitivity checks	Updated
NU 3-year post-graduate employment rate (2018–2022 average) (%)	53	Tracer study data	New (described below)
Employment rate of Higher Secondary Certificate (HSC) graduates (2017–2022 average) (%)	88.12	LFS	Updated
Wage premium by CEDP by scenario (%)	1, 2, 3	Sensitivity checks	Same as before



Item	Data/Assumption	Source	Difference with Ex Ante Analysis
Starting monthly salary of NU graduates in 2022	16,352	Tracer study data	Updated (extrapolated from 2020 and adjusted for inflation)
Start monthly salary of HSC graduates in 2022	13,162	Enterprise-based Skills Survey	Updated (extrapolated from 2010 and adjusted for inflation)
Average annual out-of-pocket expense for bachelor students in 2022	35,244	HIES	Updated (extrapolated from 2016 and adjusted for inflation)
Exchange rate (as of 31.12.22)	103.1592	Bangladesh Bank	Updated
Real interest rate (%)	5.022	World Development Indicators	Updated (average of 25 years from 1998 to 2022)
Discount rate (%)	7	Standard literature	Updated (from the ex ante assumption of 3% to be more conservative)

4. The tracer studies for the project found that the employment rate of NU college students three years after graduation was remarkably low at 53 percent (averaged over 2018 and 2022). The ex post analysis integrates this latest information into the costs and benefits analysis. While the ex ante assumed employment of graduates at the national level, this analysis extrapolates the data from the tracer study and assumes five years (at a linear rate) for NU graduates to catch up to the national employment level from three years after graduation. This approach is taken to be conservative in the IRR calculation. However, IRR calculations excluding this assumption (and maintaining the ex ante assumption of national-level employment upon graduation) are also presented.

5. It is worth noting that, remaining consistent with the methodology at appraisal, the efficiency analysis incorporates the opportunity cost in foregone income (using the annual wage of HSC graduates) and out-of-pocket expenditure (using the average annual out-of-pocket expense for bachelor students) of dropped-out students for two years. Dropped-out students are defined as the difference between cumulative beneficiary graduates with the CEDP (78 percent) and cumulative beneficiary graduates without the CEDP (75 percent). The estimated beneficiary students who would drop out in the absence of the CEDP are then projected to be 13,676. This significant negative cost is offset by the yearly positive benefits accrued from the wage premium of the project (against a wage premium of 1 percent, positive benefits are projected to initiate from year 5 after project completion), with a projection of year 15 being the turning point after which the project is estimated to yield positive net benefits. Thus, it is important to note that any positive returns from the project are long run estimates, cumulative over the assumption that college students graduate at the age of 22 and continue working until the age of 60.

6. Table 4.2 presents the results of the cost-benefit analysis based on the sensitivity analysis using three different scenarios for the rates of employment and wage premium. All calculations presented in the table are done using a 7 percent discount rate. For the low scenario, a conservative wage premium of only 1 percent and that national tertiary employment remains at the pre-project levels are assumed. Even with such conversation projections, the project still yields an IRR of 11 percent, with a positive NPV of US\$323 million. This can be treated as a lower-bound estimate, as positive externalities associated with enhanced education quality and equity arising from a healthier, better educated, and equitable/inclusive society have not been included in the analysis. Increasing the wage premium to 2 percent and 3 percent while assuming positive tertiary-level employment increases of 1 percent and 2 percent (which may



manifest well into the future, given longer-term gains from quality education), improves the project IRR up to 21 percent, well above the discount rate of 7 percent used in the calculations.

Table 4.2. Summary of Results of Cost-Benefit Analysis of the Project by Scenarios (IRR with NPV in parenthesis)

		National Tertiary Employment Rate with the CEDP		
		Low (88.8%) [no increase]	Base (89.8%) [increase of 1%]	High (90.8%) [increase of 2%]
Wage Premium	Low (1%)	11% (US\$323 million)	14% (US\$620 million)	16% (US\$917 million)
	Base (2%)	14% (US\$603 million)	17% (US\$902 million)	19% (US\$1,202 million)
	High (3%)	17% (US\$882 million)	19% (US\$1,185 million)	21% (US\$1,487 million)

7. During the project preparation phase, the ex ante analysis reported IRR estimates of 16 percent to 35 percent. While the IRR estimates reported in this analysis are lower, the conservative estimates reflect the improved assumptions made. Adhering to the ex ante assumption of graduates meeting national-level employment upon graduation yields an IRR of 13 percent with an NPV of US\$385 million (with a 7 percent discount rate, 1 percent wage premium, and pre-project-level employment). If the discount rate is relaxed further to the ex ante rate of 3 percent, the IRR increases to 17 percent with an NPV of US\$848 million.

8. It is worth noting that the recent LFS 2022 indicated a fall in the national tertiary employment rate to 88 percent. While this is likely due to factors external to the project (such as the COVID-19 pandemic impacts on the labour market), even using the reported rate of 88 percent as the post-project national tertiary employment rate, the project returns IRR estimates of 7 percent (US\$86 million), 12 percent (US\$363 million), and 15 percent (US\$640 million) for 1 percent, 2 percent, and 3 percent wage premiums, respectively (using a 7 percent discount rate).

9. The economic analysis suggests that CEDP was, ex post, a sound investment decision. Even with conservative projected increases of 1 percent in both wage premium and national tertiary employment rate, the cost-benefit analysis shows that the present value of net benefits is positive (US\$620 million), and that the program's IRR is 14 percent, which is higher than the discount rate. Given that the calculations exclude positive externalities associated with enhanced education quality for the beneficiaries, the returns are likely to be greater than the simple costs and benefits analysis presented.



ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

Dear Task Team Leader,

Thank you for sharing the draft Implementation Completion and Results Report (ICR) for the College Education Development Project (CEDP) with us. The ICR rightly captures the development objectives, implementation experiences, and results of the project. It offers valuable lessons learned for the Government of Bangladesh (GoB) and college stakeholders as we plan for the next phase of development initiatives in the sector.

The CEDP is the first development partner-supported project for the National University (NU) affiliated college sector. The NU college sector caters to more than two-thirds of all tertiary graduates in Bangladesh, being a critical sector for investments to improve the quality of higher education and workforce development in the country. The CEDP investment was timely, critical, and well designed to meet the needs of the sector.

The CEDP has successfully facilitated effective planning, governance, and management in the college sector. The project supported the development and approval of the first National Strategic Plan (NSP) for higher education colleges. The NSP has provided a consensus-based and timebound action plan for college sector development providing much needed guidance and direction to the college stakeholders for improving the quality and access to college education and college education outcomes for graduates.

The project advocated for enhancing the teaching strength in the colleges. The project offered 8,306 college teachers updated training in their fields to enhance pedagogical skills or academic qualifications. Under the project, 9,988 teachers and managers of the colleges were trained on procurement, financial management, information and communication technology (ICT), office management, recordkeeping, and monitoring and evaluation. Besides, quality assurance mechanisms were introduced in 35 colleges, which prepared self-assessments and institutional improvement plans. Additionally, 3,500 full-time college teachers were recruited and deployed to fill teacher vacancies in the colleges. The project also facilitated the amendment of legislation to support merit- and need-based recruitment of teachers in the nongovernment colleges. These important reforms are expected to have a positive impact in delivering better quality education in the colleges.

The project introduced the Institutional Development Grant (IDG) program in 120 colleges. The grants provided the much-needed facilities in the underresourced colleges through renovation and refurbishment of classrooms and establishment of ICT labs, science labs, libraries, washrooms, and mothers' corners. Moreover, the grantee colleges were also provided teaching and learning tools, that is, smart boards, desktops and laptops, photocopiers, scanners, projectors, air conditioners, and so on. Additionally, safe drinking water was ensured, power substations were set up, and campus security was ascertained in the colleges. Connectivity and campus networking were established to enhance teachers' and students' access to high-speed internet and global academic resources in the grantee colleges.

The CEDP has helped lay critical reforms and foundations for the continued development of the college sector. Eight study reports were produced under the project. These reports reveal the present status of the colleges with respect to the satisfaction of college principals, teachers, students, and employers about the teaching and learning environment created under the project and capacity strengthening of the faculty members and staff. The GoB is committed to continuing the development of the sector based on the NSP and investment plan, as the Government asserts that transformation of human resources into human capital is essential for transforming the country into a developed country in 2041.



The success of the project would not have been possible without the strong partnership and collaboration among the Bangladesh University Grants Commission, NU, Directorate of Secondary and Higher Education, National Academy of Education Management, Bangladesh Open University, University of Nottingham in Malaysia, and the CEDP Project Management Unit under the guidance of Secondary and Higher Education Division of the Ministry of Education. The concerned World Bank team has been always supportive to the Project Management Unit.

In conclusion, we would like to thank you again for your partnership and capturing the journey of CEDP. We look forward to continuing our partnership.



ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)

A. Key World Bank Documents

Official documents

World Bank. 2016. *Project Appraisal Documents on the College Education Development Project (CEDP)*. Report No. PAD1511. Washington, DC, May 11, 2016.

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Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 4* - March 4, 2018.

Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 5* - August 23, 2018.

Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 6* - February 19, 2019.

Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 7* - July 23, 2019.

Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 8* - November 14, 2019.



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Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 11* - May 5, 2021.

Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 12* - November 10, 2021.

Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 13* - May 11, 2022.

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Bangladesh - *College Education Development Project P154577 - Implementation Status Results Report: Sequence 16* - June 19, 2023.

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World Bank. 2019. *Graduate Employability of Affiliated Colleges New Evidence from Bangladesh. Graduate Tracking Survey on Affiliated Colleges of Bangladesh National University*. Report No. AUS0000633. Washington, DC. March 31, 2019.

B. Key Government Documents

Government Reports

Bangladesh. 2010. *National Education Policy 2010*. Ministry of Education.

Bangladesh. 2012. *Perspective of Plan of Bangladesh 2010–2021: Making Vision 2021 a Reality*. General Economics Division, Bangladesh Planning Commission, April 2012.

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MOE. 2022. *CEDP 11th Semi-Annual Monitoring Report. January to June 2022*. National University, Secondary and Higher Education Division, Ministry of Education, Bangladesh.

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MOE. 2023. *CEDP 13th Semi-Annual Monitoring Report. January to June 2023*. National University, Secondary and Higher Education Division, Ministry of Education, Bangladesh.

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ANNEX 7. REVISIONS TO THE RESULTS INDICATORS AND DISBURSEMENT-LINKED INDICATORS

A. Changes to Project Results Indicators

PDO and IR Is (Original)	Revised at the First Restructuring (May 1, 2019) Disbursed: US\$25.08 million	Revised at the Second Restructuring (June 14, 2021) Disbursed: US\$42.01 million	Revised at the Third Restructuring (May 10, 2023) Disbursed: US\$65.12 million	Revised at the Fourth Restructuring (December 5, 2023) Disbursed: US\$76.25 million	Rationale
PDO Indicator 1: Sector strategy and action plan developed, and the initial implementation started	Revised end target date	Revised end target DLI1 to take out the implementation of initial activities to <i>'MOE prepared an investment plan for quality enhancement of college education in line with the college subsector strategic plan'</i> .			• First restructuring: Extension of closing date for 18 months from June 2022 to December 2023 • Second restructuring: An approved investment plan based on the NSP was included given its requirement to proceed to DPP preparation and the lengthy government process and procedures for DPP approval that would go beyond project life.
PDO Indicator 2: Government college teacher vacancies filled	Revised end target date				



PDO and IR Is (Original)	Revised at the First Restructuring (May 1, 2019) Disbursed: US\$25.08 million	Revised at the Second Restructuring (June 14, 2021) Disbursed: US\$42.01 million	Revised at the Third Restructuring (May 10, 2023) Disbursed: US\$65.12 million	Revised at the Fourth Restructuring (December 5, 2023) Disbursed: US\$76.25 million	Rationale
PDO Indicator 3: Number of teachers and managers in Honours and Masters colleges trained in subject and pedagogical training under the Project	Revised end target date Revised description to include Certificate of Completion	Reduced the end target from 8,000 to 5,000 teachers and managers trained	Increased the end target from 5,000 to 6,500 teachers and managers trained		• First restructuring: Extension of closing date for 18 months. • First restructuring: Design of teacher training program with UNM (Component 2) was revised to include ECPD course for managers and teachers. • Second restructuring: COVID-19 situation that lasted for more than a year, which led to reduction of the EOP target. • Third restructuring: Increase in the EOP target following increased capacity to deliver online training.
PDO Indicator 4: Teaching and learning environment at Honours and Masters colleges upgraded (% of IDP activity milestones achieved)	Revised end target date Revised description to take out mentioning of round 2 IDG				• First restructuring: Extension of closing date for 18 months. • First restructuring: The number of IDG rounds reduced from two to one.
IRI 1: Students benefiting from direct interventions to enhance learning	• Revised end target date • Revised end target value from 2,000,000 to 2,100,000 students benefited from project intervention				• Extension of closing date for 18 months • Implementation plan revised to fit the revised closing date



PDO and IR Is (Original)	Revised at the First Restructuring (May 1, 2019) Disbursed: US\$25.08 million	Revised at the Second Restructuring (June 14, 2021) Disbursed: US\$42.01 million	Revised at the Third Restructuring (May 10, 2023) Disbursed: US\$65.12 million	Revised at the Fourth Restructuring (December 5, 2023) Disbursed: US\$76.25 million	Rationale
IRI 2: Management in the participating colleges strengthened (% of grant supported colleges publish timely annual reports) (percentage)	Revised end target date				Extension of closing date for 18 months
IRI 3: Number of colleges which completed self-assessment for quality assurance	Revised end target date				Extension of closing date for 18 months
IRI 4: Increased satisfaction levels of students and teachers of grant supported colleges about teaching and learning environment	Revised end target date				Extension of closing date for 18 months
IRI 5: PMU is fully operational	Revised end target date				Extension of closing date for 18 months
IRI 6: Midterm and follow-up tracer study conducted	Revised end target date				Extension of closing date for 18 months
IRI 7: Web-based Project Management Information System (PMIS) is established and regularly updated.	Revised end target date				Extension of closing date for 18 months



PDO and IR Is (Original)	Revised at the First Restructuring (May 1, 2019) Disbursed: US\$25.08 million	Revised at the Second Restructuring (June 14, 2021) Disbursed: US\$42.01 million	Revised at the Third Restructuring (May 10, 2023) Disbursed: US\$65.12 million	Revised at the Fourth Restructuring (December 5, 2023) Disbursed: US\$76.25 million	Rationale
IRI 8: Awareness raising activities are regularly conducted	- Revised end target date - Revised end target value from 10 to 16 awareness raising activities to be regularly conducted				- Extension of closing date for 18 months - Implementation plan revised to fit the revised closing date.

B. Changes to Project DLIs

DLIs and DLRs	Revised at the First Restructuring (May 1, 2019) Disbursed: US\$25.08 million	Revised at the Second Restructuring (June 14, 2021) Disbursed: US\$42.01 million	Revised at the Third Restructuring (May 10, 2023) Disbursed: US\$65.12 million	Revised at the Fourth Restructuring (December 5, 2023) Disbursed: US\$76.25 million	Rationale
DLI 1: Development of College Subsector Strategic Plan (US\$16 million)	Revised end target date	Revised end target DLI1 to take out the implementation of initial activities to 'MOE prepared an investment plan for quality enhancement of college education in line with the college subsector strategic plan'.			First restructuring: Extension of closing date for 18 months from June 2022 to December 2023. Second restructuring: An approved investment plan based on NSP was included given its requirement to proceed to DPP preparation and the lengthy government process and procedures for DPP approval that would go beyond project life.
DLI 2: The number of government college teacher vacancies filled (US\$18 million)					No change



DLIs and DLRs	Revised at the First Restructuring (May 1, 2019) Disbursed: US\$25.08 million	Revised at the Second Restructuring (June 14, 2021) Disbursed: US\$42.01 million	Revised at the Third Restructuring (May 10, 2023) Disbursed: US\$65.12 million	Revised at the Fourth Restructuring (December 5, 2023) Disbursed: US\$76.25 million	Rationale
DLI 3: Improvement of teacher management system for non-government colleges (US\$9 million)	Revised end target date		Dropped DLR Year 6 on establishment of NTSC and added DLR Year 7 on 'MOE has placed the draft Amendment of NTRCA Act to Cabinet Division for approval'.		• First restructuring: Extension of closing date for 18 months. • Third restructuring: Government decided to pursue the reform in nongovernment teacher recruitment through Amendment of the NTRCA Act 2005.
DLI 4: Improved teaching and learning environment in selected government and non-government colleges (US\$25 million)	Revised end target date Revised description to take out mentioning of round 2 IDG	Reset DLI4's Year 6 target to 65% (instead of 75%)			• First restructuring: Extension of closing date for 18 months. • Second restructuring: The number of IDG rounds reduced from two to one.
DLI 5: Number of teachers and managers in Honours and Masters colleges trained in subject, pedagogical, and managerial trainings (US\$18 million)	Revised end target date Revised verification protocol to include Certificate of Completion	Reduced the end target from 8,000 to 5,000 teachers and managers trained	Increased the end target from 5,000 to 6,500 teachers and managers trained		• First restructuring: Extension of closing date by 18 months. • Second restructuring: COVID-19 situation that lasted for more than a year, which led to reduction of final target for teachers and managers trained. • Third restructuring: Increase in the final target following increased capacity to deliver online training.



DLIs and DLRs	Revised at the First Restructuring (May 1, 2019) Disbursed: US\$25.08 million	Revised at the Second Restructuring (June 14, 2021) Disbursed: US\$42.01 million	Revised at the Third Restructuring (May 10, 2023) Disbursed: US\$65.12 million	Revised at the Fourth Restructuring (December 5, 2023) Disbursed: US\$76.25 million	Rationale
DLI 6 (New): Establishment of Learning Management Information System and content development (US\$6 million)		Added new DLI 6 on the establishment of an LMIS and content development		Dropped DLI 6 Year 6 (b) on content development for LMIS	• Second restructuring: This new COVID-19 response DLR in Year 5 supported strengthening of the college sector's resilience and capacity to deliver online/blended learning programs for college students. • Fourth restructuring: The e-content activity was dropped due to delay in onboarding technical expertise and the NU's decision to develop e-content with its own resources under its broader college development program.



ANNEX 8. DESCRIPTION OF PROJECT COMPONENTS

1. The CEDP consisted of three components: (a) Components 1 and 2 are results-based components and (b) Component 3 provided technical assistance. The descriptions are derived from the Financing Agreement and the PAD.
2. **Component 1: Strengthening Strategic Planning and Management Capacity (Estimated Cost: US\$48 million; Actual Cost: US\$48 million).** The objective of this component was to address planning, governance, and management challenges and place reforms at the system level.
3. **Subcomponent 1.1: Development of a Strategic Plan for the College Subsector.** The objective of this subcomponent was to develop a strategic plan for the college subsector to provide directions and goals for the college education subsector and a consensus-based plan of action. This subcomponent was supported by DLI 1: Development of College Subsector Strategic Plan.
4. **Subcomponent 1.2: Improvement of the Teacher Management System.** The objective of this subcomponent was to support the MOE's efforts to (a) fill teacher vacancies in government colleges and (b) reform the teacher recruitment procedures for nongovernment colleges. This subcomponent was supported by the following two DLIs: DLI 2: The number of government college teacher vacancies filled and DLI 3: Improvement of teacher management system for nongovernment colleges.
5. **Component 2: Improving Teaching and Learning Environment in Participating Colleges (Estimated Cost: US\$42 million; Actual: US\$42 million).** This component's objective was to address the challenge of providing quality and relevant college education at the institutional level.
6. **Subcomponent 2.1: Improving Teaching-Learning Environment in Colleges.** The objective of this subcomponent was to support the development of IDPs and its implementation by eligible participating government and nongovernment colleges. Around 120 colleges received IDGs to enhance the teaching-learning environment through the implementation of quality and management improvement activities. These activities included upgrading basic teaching-learning facilities and internet connectivity, enhancing the market relevance of college education through the development of students' soft skills and industry/employer links, introducing quality assurance measures, and strengthening management capacity along with upgrading fiduciary systems in the colleges. This subcomponent was supported by DLI 4: Improved teaching and learning environment in selected government and nongovernment colleges.
7. **Subcomponent 2.2: Strengthening Teaching and Management Capacity in Colleges.** The objective of this subcomponent was to bring about qualitative improvements in teaching and management in the college subsector by supporting (a) the training of trainers; (b) teacher training; (c) capacity building of college principals; and (d) capacity building of policy makers, leaders, and managers for subsector development. This subcomponent was supported by the following DLI: DLI 5: Number of teachers and managers in Honours and Masters colleges trained in subject, pedagogical, and managerial trainings.
8. **Component 3: Project Management, Communication, and Monitoring and Evaluation (Estimated Cost: US\$10 million; Actual Cost: US\$4.12 million).** The objective of this component was to support project management, communication, and M&E. It supported the establishment of a PMU, specific capacity building, and technical assistance necessary for project implementation and several analytical studies.



ANNEX 9. SUMMARY ON INSTITUTIONAL DEVELOPMENT GRANTS

1. The main objective of IDG under the CEDP was to assist the participating colleges in establishing enabling conditions and environment to improve the quality and relevance of teaching and learning and to enhance the institutional management practices. The IDG program addressed many challenges faced by the colleges in raising the quality and relevance of higher education offered by the NU. These included (a) inadequate teaching and learning infrastructure, especially for science and technology disciplines; (b) lack of relevance with the industry and service sector jobs; (c) absence of quality assurance mechanisms at the institutional and subsector levels; and (d) inadequate planning and management capacity for institutional development.

2. The specific objectives of the IDG include (a) enhancement of the teaching skills and academic knowledge of college teachers for providing high-quality teaching; (b) modernization of the teaching and learning facilities and equipment through renovation and refurbishment of lecture rooms, laboratories, and libraries and procurement of scientific instruments, multimedia, and other teaching tools; (c) introduction of the institution-based quality assurance mechanisms; (d) establishment of campus networking and internet connectivity through the BdREN and other network services; (e) development of soft skills programs; (f) introduction and enhancement of institution-industry links; and (g) strengthening of planning and management capacity for institutional development.

II. IDG Selection Process

3. The IDGs were offered to NU-accredited Honours and Masters courses in government colleges, nongovernment colleges, and girls' colleges.

- The eligibility criteria for government colleges were the following: (a) the colleges must have at least three batches of Honours courses graduated; (b) total annual intake capacity in Honours/Masters courses must be greater than 1,000; and (c) the audited financial statements not older than the previous two years must be available.
- The eligibility criteria for nongovernment colleges were the following: (a) at least three batches of Honours courses graduated; (b) the total annual intake capacity in Honours/Masters courses should be greater than 500; and (c) the audited financial statement not older than the previous two years must be available.
- The eligibility criteria for the girls' colleges were the following: (a) must have at least three batches of Honours courses graduated; (b) the total annual intake capacity in Honours/Masters courses is greater than 120; and (c) the audited financial statement not older than the previous two years must be available.

4. The colleges were then grouped into three categories (A, B, and C) for receiving the IDG to reflect the diversity of the academic and institutional capacity of colleges, such as size of enrollment, physical facilities, teaching personnel, subjects and courses offered, and other facilities, and so on. This was to ensure a level playing field for all the eligible colleges and equitable distribution of the grant among these colleges. Table 9.1 presents the number of colleges for each group.

Table 9.1. Criteria for Grouping of Colleges

College Groups	Government Colleges (number of students)	Nongovernment Colleges (number of students)	Total Number of Colleges
A	5,000+	1,000+	20



College Groups	Government Colleges (number of students)	Nongovernment Colleges (number of students)	Total Number of Colleges
B	1,000–5,000	500–1,000	68
C (Girls' Colleges)	120	120	34
Total			122

III. IDG Fund and Allocation

5. Total amount provided for the IDG was BDT 50,214.85 lakh (approximately US\$62.7 million). The provision was made to allocate this amount to colleges in two rounds of IDGs, but it was later fully allocated in one round of IDG while maintaining the target number of colleges. The IDG awards were distributed, with 60 percent allocated to government colleges and 40 percent to nongovernment colleges in each group. The ceiling per institution for each group was as follows: Group A - BDT 800 lakh (US\$1 million); Group B - BDT 400 lakh (US\$500,000); and Group C - BDT 240 lakh (US\$250,000). It is to be noted that 50% of grants allocated to each group were provided to the colleges located in urban periphery and rural areas.

IV. Outputs

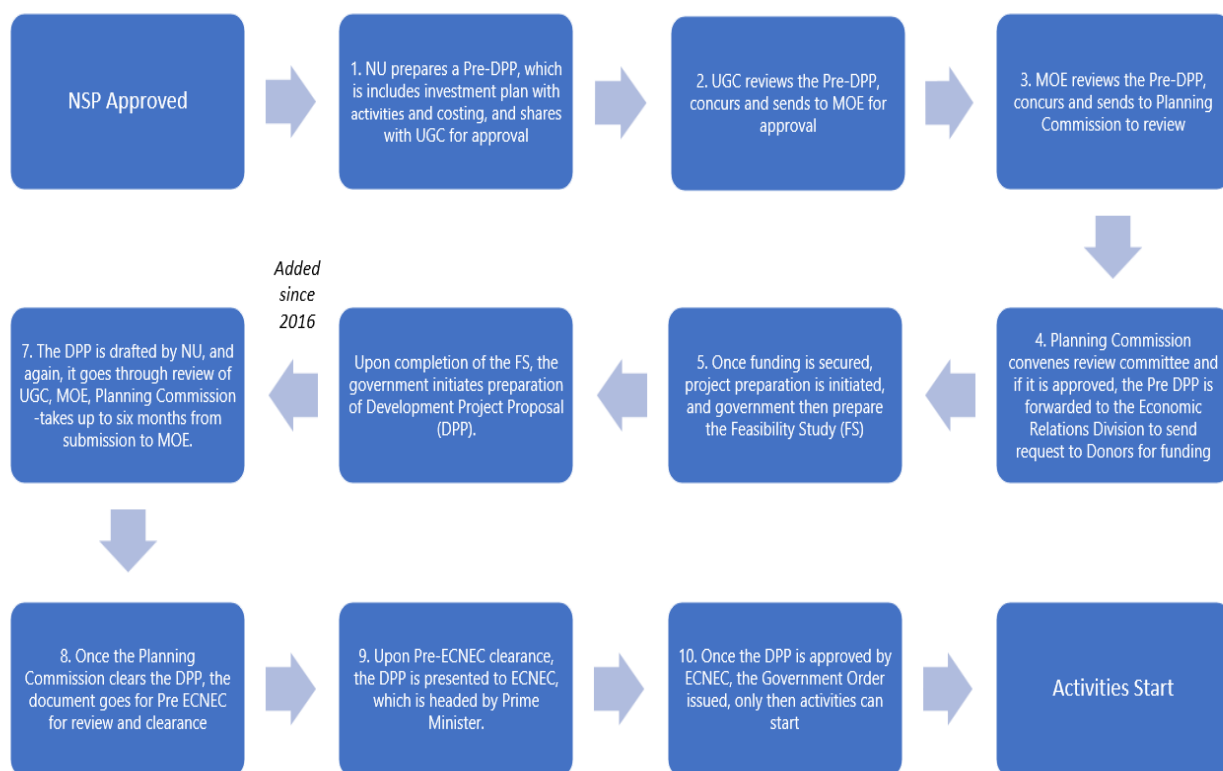
6. In compliance with objectives of the IDG provision, the participating colleges formulated IDPs to set out infrastructure development goals and actions necessary to achieve the stated goals and indicators to measure achievements within the three-year implementation period. The IDPs outline plans for procurement activities and financial management to be undertaken by the IDP subproject. The IDG funds provided to the subprojects support implementation of the IDP in the awarded institutions. The colleges completed the following: renovation and refurbishment in 1,889 classrooms, 286 science labs, 158 ICT labs, 1,166 washrooms, 62 mothers' corners, and 624 libraries, among others. The IDGs provided ICT facilities, such as over 9,000 computers, 1,002 digital smartboards, and 125 digital noticeboards; safe drinking water through 1,121 purifiers; over 20,000 lab equipment and items for science labs; and 938,653 furniture and fixtures. The IDGs were used to connect 110 colleges to BdREN and establish 94 campus networks for high-speed internet access for students and teachers. In addition, 35 colleges used IDGs to conduct self-assessments, which enabled them to review, assess, and plan for improving the teaching-learning environment and teaching quality.

7. The IDGs also entailed increasing management capacity of over 9,988 teachers and managers through training and workshops, in areas such as procurement, financial management, ICT, office management, recordkeeping, and M&E.

8. As of 2023, the total beneficiary students in IDG colleges were 757,034. Among them 593,217 were undergraduates and 150,295 were graduates. There were 7,811 faculty members who benefitted from the IDG intervention in the colleges.



ANNEX 10. GOVERNMENT'S APPROVAL STEPS OF INVESTMENT PLAN



Source: GoB. October 2016. Government Development Project Submission, Processing, Approval, and Revision Method. Planning Commission: Dhaka.

**ANNEX 11. TRAINING COURSES AND PARTICIPANTS UNDER SUBCOMPONENT 2.2**

Sl. No.	Type of Training	Total	Male	Female
1.	Master's in Education (Master Trainer) for College Teachers	453	403	50
2.	Master's in Educational Leadership Management (MA in ELM) for Principals	32	28	4
3.	Leaders (ECPD)	79	72	07
4.	Future leader (ECPD)	600	543	57
5.	Teachers training (ECPD)	2,194	1,876	318
6.	Subject-Based teachers training at NU (online/blended)	3,226	2,444	782
7.	Subject-based teachers training at NU (face-to-face)	1,689	1343	346
8.	Policy makers	33	30	3
Total teachers and managers in Honours and Masters colleges trained in subject and pedagogical training under the Project (as of December 31, 2023)		8,306	6,739	1,567



ANNEX 12. SUMMARY OF STUDIES AND EVALUATION UNDER CEDP

1. **Baseline satisfaction survey.** The study conducted by the BIDS, Dhaka, from February 25 to October 15, 2019, aimed to assess satisfaction levels among principals, teachers, students, and employers regarding the teaching and learning environment in IDG-awarded NU-affiliated Honours and Masters level tertiary colleges. It measured baseline satisfaction levels among college stakeholders across a representative sample of 75 government and nongovernment colleges in all eight Bangladeshi divisions, consisting of 44 government colleges and 31 nongovernment colleges. The study employed both quantitative and qualitative methods, including structured questionnaires and focus group discussions (FGDs), to gather data from students, teachers, principals, and potential employers of NU graduates.

2. The study's major findings indicate satisfaction levels on a Likert Scale (with 1 = Not Satisfied and 5 = Highly Satisfied) as follows: principals 2.56, teachers 2.42, students 2.59, and employers 3.78. Based on these findings, the study proposes several recommendations:

- Implement English language programs to enhance communication skills.
- Encourage one-on-one counseling between teachers and students.
- Establish mentoring programs for junior teachers by senior teachers.
- Provide pedagogical and on-the-job training for newly recruited teachers.
- Foster collaboration with prospective employers and offer internship opportunities for students.
- Develop technical skills from an employer's perspective.
- Establish college alumni associations.
- Increase the number of computers and demonstrators in college laboratories.
- Address the need for more teachers to fill current vacancies and ensure proper training for newly recruited teachers.
- Arrange short courses, including soft skill development programs.
- Organize industry collaboration through job fairs, preferably at the district level, to facilitate employment opportunities.
- Allocate funds for upgrading labs, research facilities, and IT infrastructure as needed.

3. **Tracer study on graduates of tertiary-level colleges - Phase 1.** The BIDS conducted a study from February 6 to July 5, 2021, aiming to track the NU graduates from a sample of NU-affiliated tertiary colleges. Employing both quantitative and qualitative methods, the study utilized four sets of questionnaires targeting principals, teachers, students, and employers. The survey encompassed graduates, current students, institution heads/teachers, and employers across 61 colleges, with 17 being government and 44 nongovernment institutions.

4. The key findings of the study are as follows:

- 21.0 percent of graduates are employed full-time, while 66.0 percent are unemployed, and 1.5 percent are self-employed.
- 7 percent are pursuing further studies, and 5 percent are not employed.



- Paid employment rates are 11.64 percent for females and 30.74 percent for males.
5. The report suggests several recommendations to improve job opportunities for the NU graduates, including the following:
- Organizing frequent job fairs in collaboration with industries
 - Increasing investments in higher education institutes, focusing on online learning solutions such as videoconferencing
 - Providing ICT knowledge to teachers for better student guidance
 - Offering more scholarships to attract students to STEM¹⁰ subjects with labor market relevance
 - Urgently addressing teacher recruitment and training policies
 - Regularly updating course curricula to enhance learning
 - Enhancing the job market relevance of education
 - Arranging short courses for various skills, including soft skill development
 - Organizing annual job fairs, preferably at the district level, to foster industry collaboration
 - Allocating funds to upgrade labs, research facilities, and IT environments to facilitate online-based hybrid learning
6. The 'Effectiveness and Situation Assessment of Teachers' Training under CEDP' was conducted by Samahar Consultants Ltd. in Dhaka from January 1 to June 9, 2022. The study aimed to evaluate the midterm performance of teacher training programs under the CEDP. It employed a methodology involving the selection of 30 percent of total training recipient teachers from five types of training, including master trainers, teachers training, ECPD, future leader training (ECPD), and leaders Training (ECPD) under the UNM, and subject-based training under the NU, totaling 988 sample trainees out of 3,293 nationwide.
7. A multi-stage purposive sampling method was used to select 76 colleges representing all districts, with teacher respondents distributed across divisions as follows: Dhaka (305), Chottogram (143), Rajshahi (98), Khulna (143), Sylhet (63), Barishal (75), Rangpur (89), and Mymensingh (72). The study employed both quantitative and qualitative surveys, including respondent questionnaires, key informant interviews (KIIs), and FGDs, totaling 32 KIIs and 17 FGDs.
8. Analysis revealed positive responses from respondents across all training programs, indicating improvements in skills, attitude, knowledge, teacher-student relationships, ICT use, pedagogy/andragogy concepts, teaching-learning skills, classroom management, subject-based discussions, lesson plan preparation, teaching punctuality, and professionalism.
9. Recommendations proposed in the study report include the following:
- Ensuring leave or deputation for teachers to pursue MA in Education
 - Providing IELTS practice sessions
 - Mandating MA in Education training for all NU college teachers

¹⁰ STEM = Science, technology, engineering, and mathematics.



- Providing soft loans for purchasing ICT devices
- Conducting follow-up training every two to three years.

10. The midline satisfaction survey, conducted by the BIDS from March 15 to October 31, 2022, targeted 10 percent of the population, resulting in a sample of 75 Honours and Masters colleges affiliated with the NU. Covering 41 districts, the study utilized the same sample selection methodology as the baseline satisfaction survey, comprising 60 percent government colleges and 40 percent nongovernment colleges.

11. Structured questionnaires were administered to current students, teachers, principals, and potential employers of NU graduates. The BIDS surveyed 3,017 students, 1,245 teachers, 73 principals, and 196 employers. Qualitative approaches included 8 FGDs with students, 8 FGDs with teachers, and 15 key informant interviews with employers.

12. The study aimed to assess satisfaction levels among principals, teachers, students, and employers regarding the teaching and learning environment. Key findings indicated satisfaction levels on a five-point Likert Scale as follows: principals 2.75, teachers 2.55, students 2.71, and employers 3.73. Principals' satisfaction increased significantly, while teachers' satisfaction decreased. Students' satisfaction improved compared to the baseline, while employers' satisfaction remained stable. It is to be noted that prolonged college closure from COVID-19 affected the satisfaction level of students and teachers during the midterm survey.

13. Recommendations from the report include the following:

- Organizing annual job fairs at the district level to foster industry collaboration
- Introducing short courses to enhance job market opportunities for NU-affiliated colleges
- Providing foreign training for NU teachers
- Arranging training before teacher recruitment
- Offering long-term subject-based and pedagogical training for quality teaching
- Enhancing collaboration between NU-affiliated colleges and universities
- Establishing and activating alumni associations in NU-affiliated colleges.

14. **Follow-up tracer study on graduates of tertiary-level colleges - (Tracer Study -2).** The BIDS, Dhaka, conducted a study from February 9 to October 15, 2023, aiming to assess the labor market relevance of tertiary college education over time. This survey builds upon the previous tracer study on the NU graduates and adopts a mixed-method approach combining quantitative and qualitative methods. The sample includes college principals, students, graduates, and current employers of NU graduates, with structured questionnaires administered to each group. Overall, the study reveals a decrease in the unemployment rate of NU graduates from 66 percent in 2021 to 48 percent in 2023.

15. Key findings of the study include the following:

- Private college graduates demonstrate relatively higher success in the labor market.
- A significant portion of graduates remain unemployed, particularly those from BA (pass) courses, political science, library management, Bengali, and Islamic history, and culture.



- Graduates of English, economics, accountancy, sociology, finance, and banking exhibit lower unemployment rates.
- Employers prioritize soft skills, ICT, communication, and language skills.

16. Recommendations for improving the teaching and learning environment of NU-affiliated colleges include the following:

- Enhancing ICT usage and providing ICT skills training
- Implementing periodic institution-level graduate tracking
- Establishing job placement support services and career counseling within colleges
- Organizing annual job fairs at the district level to foster industry collaboration
- Encouraging enrollment in courses with higher market demand, such as economics, accounting, English, finance, and banking
- Introducing short course facilities to increase job market opportunities
- Enhancing English language and communication skills of college graduates
- Strengthening alumni associations to facilitate networking and job-seeking support for current students.

17. **The study on project effectiveness including measurement of endline satisfaction survey.** The BIDS conducted a study from April 15 to December 15, 2023, aiming to assess the effectiveness of project activities and measure endline satisfaction levels of beneficiaries (students, teachers, and employers) regarding teaching quality and relevance. Employing a mixed-method approach, the study sampled IDG-awarded colleges (treatment group) and IDG non-recipient colleges (control group) based on baseline satisfaction survey data.

18. Separate questionnaires were designed for students, teachers, college principals, and current employers of NU graduates, alongside qualitative approaches such as FGDs and KIIs among teachers. Survey instruments were pilot tested in selected colleges, and consistent sampling procedures from previous baseline and midterm studies were maintained.

19. Key findings include satisfaction levels on a five-point Likert Scale: principals: 2.59, teachers: 2.68, students: 3.53, and employers: 3.66. The highest mean satisfaction for students was observed for teaching-learning facilities, soft skill development, academic infrastructure, industrial cooperation, and internet connectivity. Overall satisfaction rates increased from baseline to endline for students and teachers, with employers' satisfaction remaining stable.

20. The study highlights increased use of management information systems, internet connectivity, and Wi-Fi networks continue to support academic activities.

21. Recommendations proposed in the report include the following:

- Undertake more higher education development and expand project coverage to additional colleges.
- Continue regular teachers' training across all colleges, with clear guidelines for teacher selection.



- Recruit more teachers, especially in nongovernment colleges, to maintain quality teaching standards.
- Consider students' evaluation of teachers to enhance teaching quality.
- Continue satisfaction surveys to monitor improvements and ensure clear exit plans for project sustainability.
- Implement periodic progress tracking, tracer studies, and satisfaction surveys irrespective of specific projects.



ANNEX 13. COLLEGE SECTOR ENROLLMENT TRENDS, 2016–2023

	2016	2017	2018	2019	2022	2023
Total Enrollment	1,600,000	1,700,000	1,800,000	2,000,000	2,227,381	2,342,802
Female Enrollment (% share of total)	7,52,000 (47%)	7,48,000 (44%)	8,28,000 (46%)	9,29,000 (47%)	1,076,937 (48%)	1,124,924 (48%)
Total GER (Tertiary)	17	19	20	23	23	n.a.
Female GER (Tertiary)	14	15	16	19	21	n.a.

Source: BANBEIS Annual Reports, various years, for college enrollments; World Bank Databank for GER data

Note: Data are provided for available years.